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		Questions Sheet	1 - 3	4	5	6	7	TOTAL
Student Name and Surname								
Student ID								
Course	Optimization Techniques - Midterm					Date	15.04.2026	
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1. 10p. Using FONC, how many stationary points does $f(x, y) = x^4 + y^4 - 4x^2 - 2y^2 + 3$ have?

a. 3

b. 4

c. 6

☒ d. 9

e. 12

$$f = x^4 + y^4 - 4x^2 - 2y^2 + 3$$

$$\nabla f = \begin{bmatrix} 4x^3 - 8x \\ 4y^3 - 4y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$3 \times 3 = 9$ candidate points

$$(0, 0), (0, 1), (0, -1)$$

$$(\sqrt{2}, 0), (\sqrt{2}, 1), (\sqrt{2}, -1)$$

$$(-\sqrt{2}, 0), (-\sqrt{2}, 1), (-\sqrt{2}, -1)$$

$$4x(x^2 - 2) = 0$$

$$x = 0 \quad x = \pm\sqrt{2}$$

$$4y(y^2 - 1) = 0$$

$$y = 0 \quad y = \pm 1$$

2. 10p. Consider the matrix: $A = \begin{bmatrix} -a & 2 & 0 \\ 2 & -3 & 1 \\ 0 & 1 & -2 \end{bmatrix}$. For which of the following values of a is A negative definite?

a. $a > 0$

b. $a > \frac{4}{3}$

c. $a > \frac{8}{5}$

d. $a > 2$

e. $a > 3$

LPM

$$\begin{bmatrix} -a & 2 & 0 \\ 2 & -3 & 1 \\ 0 & 1 & -2 \end{bmatrix}$$

$$\Delta_1 = -a < 0, a > 0$$

$$\Delta_2 = \begin{vmatrix} -a & 2 \\ 2 & -3 \end{vmatrix} = 3a - 4 > 0, a > 4/3$$

$$\Delta_3 = (-1)^{2+3} \begin{vmatrix} -a & 2 \\ 0 & 1 \end{vmatrix} - 2 \cdot (-1)^{3+3} \begin{vmatrix} -a & 2 \\ 2 & -3 \end{vmatrix}$$

$$= -(-a - 0) - 2(3a - 4)$$

$$= a - 6a + 8 = 8 - 5a < 0$$

$$8 < 5a$$

$$a > \frac{8}{5}$$

$$g^1 = \begin{pmatrix} -2 \\ -1 \end{pmatrix} - \begin{pmatrix} -4 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$[4y + x - 2]$$

$$\nabla f(x^{(0)}) = \begin{pmatrix} -4 \\ -2 \end{pmatrix}$$

Corrective term.

$$\nabla f(x^{(1)}) = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$$

$$\frac{\begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix}}{(1, 0) \begin{pmatrix} 2 \\ 1 \end{pmatrix}} = \frac{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}}{2} = \begin{pmatrix} 1/2 & 0 \\ 0 & 0 \end{pmatrix}$$

Damping term.

$$\frac{\begin{pmatrix} 2 \\ 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \end{pmatrix}}{(2 \ 1) \begin{pmatrix} 2 \\ 1 \end{pmatrix}} = \frac{\begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}}{4+1} = \begin{pmatrix} \frac{4}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} \end{pmatrix}$$

$$H^{(1)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{pmatrix} 1/2 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 4/5 & 2/5 \\ 2/5 & 1/5 \end{pmatrix} = \begin{pmatrix} 7/10 & -2/5 \\ -2/5 & 4/5 \end{pmatrix}$$

3. 10p. Which of the following matrices are positive semidefinite?

$$A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- a. Only A_1 b. Only A_3 c. A_1 and A_2 **(d.) A_1 and A_3** e. A_1 , A_2 , and A_3

LPMs
 $A_1 \Rightarrow \Delta_1 = 1$
 $\Delta_2 = 2$
 $\Delta_3 = 0$

PMs
 1st order: 1, 2, 0
 2nd: $\begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} = 2$, $\begin{vmatrix} 2 & 0 \\ 0 & 0 \end{vmatrix} = 0$, $\begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} = 0$ **PSD**
 3rd = 0

$A_2 \Rightarrow$ LPMs
 $\Delta_1 = 1 > 0$
 $\Delta_2 = 1 - 4 < 0$ **indefinite**

$A_3 \Rightarrow$ LPMs
 $\Delta_1 = 2 > 0$
 $\Delta_2 = 4 - 1 = 3 > 0$
 $\Delta_3 = 0$ **PSD.**

PMs
 1st: 2, 2, 0
 2nd: $\begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 3$, $\begin{vmatrix} 2 & 0 \\ 0 & 0 \end{vmatrix} = 0$, $\begin{vmatrix} 2 & 0 \\ 0 & 0 \end{vmatrix} = 0$
 3rd = 0.

4. Consider the quadratic function $f(x) = \frac{1}{2}x^T Q x - b^T x$ where $Q = \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$.

i. 5p. For $x^{(0)} = \begin{bmatrix} 0 & 0 \end{bmatrix}^T$, what is the step size α_0 of the steepest descent method with exact line search?

a. $1/4$

☒ b. $1/3$

c. $1/2$

d. $2/3$

e. $3/4$

ii. 10p. What is the next iterate $x^{(1)}$?

a. $\begin{bmatrix} \frac{4}{3} & \frac{4}{3} \end{bmatrix}^T$

b. $\begin{bmatrix} \frac{3}{4} & 0 \end{bmatrix}^T$

c. $\begin{bmatrix} 0 & \frac{4}{3} \end{bmatrix}^T$

☒ d. $\begin{bmatrix} \frac{4}{3} & 0 \end{bmatrix}^T$

e. $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$

iii. 10p. What is the next iterate $x^{(2)}$?

☒ a. $\begin{bmatrix} \frac{4}{3} & \frac{4}{9} \end{bmatrix}^T$

b. $\begin{bmatrix} \frac{3}{2} & \frac{1}{2} \end{bmatrix}^T$

c. $\begin{bmatrix} \frac{8}{9} & \frac{4}{9} \end{bmatrix}^T$

d. $\begin{bmatrix} \frac{4}{9} & \frac{4}{3} \end{bmatrix}^T$

e. $\begin{bmatrix} 1 & \frac{1}{3} \end{bmatrix}^T$

$$\frac{1}{2} \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$
$$\begin{bmatrix} 3x - y & -x + 3y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\frac{1}{2} (3x^2 - xy - xy + 3y^2) - 4x = \frac{3}{2}x^2 - xy + \frac{3}{2}y^2 - 4x = f$$

$$\frac{1}{2} [x \ y] \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} - [x \ y] \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

$$[3x - y \quad -x + 3y] \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\frac{1}{2} (3x^2 - xy - xy + 3y^2) - 4x = \frac{3}{2}x^2 - xy + \frac{3}{2}y^2 - 4x = f$$

$$\nabla f = \begin{bmatrix} 3x - y - 4 \\ -x + 3y \end{bmatrix} \quad \nabla f \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 0 \end{pmatrix}$$

$$x^{(1)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} -4 \\ 0 \end{pmatrix} = \begin{pmatrix} 4\alpha \\ 0 \end{pmatrix}$$

$$f \begin{pmatrix} 4\alpha \\ 0 \end{pmatrix} = \frac{3}{2} \cdot 16\alpha^2 - 4 \cdot 4\alpha = 24\alpha^2 - 16\alpha = \phi(\alpha)$$

$$\phi'(\alpha) = 48\alpha - 16 = 0 \quad \alpha = 1/3$$

$$x^{(1)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} -4 \\ 0 \end{pmatrix} = \begin{pmatrix} 4/3 \\ 0 \end{pmatrix} \quad \nabla f(x') = \begin{pmatrix} 0 \\ -4/3 \end{pmatrix}$$

$$x^{(2)} = \begin{pmatrix} 4/3 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 0 \\ -4/3 \end{pmatrix} = \begin{pmatrix} 4/3 \\ 4\alpha/3 \end{pmatrix} = \frac{4}{3} \begin{pmatrix} 1 \\ \alpha \end{pmatrix}$$

$$f(x^{(2)}) = \frac{3}{2} \cdot \frac{16}{9} - \frac{16}{9} \alpha + \frac{3}{2} \cdot \frac{16}{9} \alpha^2 - 4 \cdot \frac{4}{3} = \phi(\alpha)$$

$$\phi'(\alpha) = -\frac{16}{9} + \frac{48}{9} \alpha = 0 \quad \alpha = \frac{16}{48} = \frac{1}{3}$$

$$x^{(2)} = \begin{pmatrix} 4/3 \\ 0 \end{pmatrix} - \frac{1}{3} \begin{pmatrix} 0 \\ -4/3 \end{pmatrix} = \begin{pmatrix} 4/3 \\ 4/9 \end{pmatrix}$$

5. Consider the quadratic function $f(x, y) = 3x^2 + 2y^2 + 2xy - 4x - 6y$ with initial point $x^{(0)} = [1 \ 1]^T$.

i. 5p. What is the gradient $\nabla f(x^{(0)})$?

☒ a. $[4 \ 0]^T$

b. $[0 \ 4]^T$

c. $[2 \ 2]^T$

d. $[-4 \ 0]^T$

e. $[0 \ 0]^T$

ii. 5p. What is the Hessian matrix $\nabla^2 f(x^{(0)})$?

☒ a. $\begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix}$

b. $\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$

c. $\begin{bmatrix} 6 & 0 \\ 0 & 4 \end{bmatrix}$

d. $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$

e. $\begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix}$

iii. 5p. What is the Newton direction $p^{(0)}$ obtained by solving $\nabla^2 f(x^{(0)}) p^{(0)} = -\nabla f(x^{(0)})$?

- a. $\left[-\frac{4}{5} \quad \frac{2}{5}\right]^T$ b. $\left[\frac{4}{5} \quad -\frac{2}{5}\right]^T$ c. $\left[-\frac{2}{5} \quad \frac{4}{5}\right]^T$ d. $\left[\frac{2}{5} \quad -\frac{4}{5}\right]^T$ e. $\left[-1 \quad 1\right]^T$

iv. 5p. What is the Newton iterate $x^{(1)}$?

- a. $\left[\frac{1}{5} \quad \frac{7}{5}\right]^T$ b. $\left[\frac{2}{5} \quad \frac{6}{5}\right]^T$ c. $\left[\frac{3}{5} \quad \frac{4}{5}\right]^T$ d. $\left[\frac{1}{5} \quad \frac{6}{5}\right]^T$ e. $\left[1 \quad 1\right]^T$

$$f = 3x^2 + 2y^2 + 2xy - 4x - 6y$$

$$x^{(0)} = (1, 1)^T$$

$$\nabla f = \begin{bmatrix} 6x + 2y - 4 \\ 4y + 2x - 6 \end{bmatrix} \quad \nabla^2 f = \begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix}$$

$$\nabla f(x^{(0)}) = \begin{bmatrix} 4 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix} \underbrace{\begin{bmatrix} a \\ b \end{bmatrix}}_{p^{(0)}} = - \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

$$6a + 2b = -4$$

$$2a + 4b = 0$$

$$a = -2b$$

$$-12b + 2b = -4$$

$$b = \frac{4}{10} = \frac{2}{5}$$

$$a = -\frac{4}{5}$$

$$x^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} -4/5 \\ 2/5 \end{pmatrix} = \begin{pmatrix} 1/5 \\ 7/5 \end{pmatrix}$$

6. Consider $f(x, y) = e^x - 4x + \frac{1}{2}y^2$ with initial point $x^{(0)} = [0 \ 1]^T$, and suppose backtracking line search is used with $\rho = \frac{1}{2}$ and $c = \frac{1}{4}$.

$$x^{(k+1)} = x^{(k)} + \alpha_k p^{(k)}, \quad \nabla^2 f(x^{(k)}) p^{(k)} = -\nabla f(x^{(k)}), \quad f(x^{(k)} + \alpha p^{(k)}) \leq f(x^{(k)}) + c \alpha \nabla f(x^{(k)})^T p^{(k)}$$

i. 5p. Which trial step size is accepted at the first iteration of the damped Newton method?

a. 1

☒ b. $1/2$

c. $1/4$

d. $1/8$

e. $1/16$



ii. 5p. What is the value of the right-hand side of the Armijo condition corresponding to the accepted step size?

a. $-\frac{1}{2}$

☒ b. $\frac{1}{4}$

b. $\frac{1}{2}$

c. $\frac{3}{4}$

d. 1

iii. 5p. What is the accepted iterate $x^{(1)}$?

a. $[3 \ 0]^T$

☒ b. $[\frac{3}{2} \ \frac{1}{2}]^T$

b. $[\frac{3}{4} \ \frac{3}{4}]^T$

c. $[\frac{1}{2} \ \frac{3}{2}]^T$

d. $[0 \ 1]^T$

$$f(x, y) = e^x - 4x + \frac{1}{2}y^2, \quad x^{(0)} = (0, 1)^T \quad p = \frac{1}{2} \quad c = \frac{1}{4}$$

$$\nabla f = \begin{bmatrix} e^x - 4 \\ y \end{bmatrix} \quad \nabla^2 f = \begin{bmatrix} e^x & 0 \\ 0 & 1 \end{bmatrix}$$

$$\nabla^2 f(x^0) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \nabla f(x^0) = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix} \quad \text{where } p^{(0)} = \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\begin{aligned} a &= 3 \\ b &= -1 \end{aligned}$$

$$p^{(0)} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

$$f(x^{(0)}) = 1 - 0 + \frac{1}{2} = \frac{3}{2}$$

Armijo: $\alpha = 1$

Armtjo: $\boxed{\alpha=1}$

$$\text{LHS: } f\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} + 1 \cdot \begin{pmatrix} 3 \\ -1 \end{pmatrix}\right) = f\left(\begin{pmatrix} 3 \\ 0 \end{pmatrix}\right) = e^3 - 12$$

$$\text{RHS: } \frac{3}{2} + \frac{1}{4} \cdot 1 \cdot (-3 \quad 1) \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \frac{3}{2} + \frac{1}{4} (-9 - 1) = \frac{3}{2} - \frac{5}{2} = -1$$

$$e^3 - 12 \stackrel{?}{\leq} -1 \quad \times \quad \alpha=1 \text{ is rejected.}$$

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$\boxed{\alpha=1/2}$

$$\text{LHS: } f\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 3 \\ -1 \end{pmatrix}\right) = f\left(\begin{pmatrix} 3/2 \\ 1/2 \end{pmatrix}\right) = e^{3/2} - 4 \cdot \frac{3}{2} + \frac{1}{2} \cdot \frac{1}{4}$$

$$= 4.48 - 6 + \frac{1}{8}$$

$$\text{RHS} = \frac{3}{2} + \frac{1}{4} \cdot \frac{1}{2} \cdot (-\frac{5}{10}) = \frac{1}{4}$$

$$= -1.3933$$

$$-1.3933 \stackrel{?}{\leq} \frac{1}{4} \quad \checkmark \quad \alpha=\frac{1}{2} \text{ accepted.}$$

$$X^{(1)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 3/2 \\ 1/2 \end{pmatrix}$$

7. 10p. Consider the quadratic function $f(x, y) = x^2 + 2y^2 + xy - 4x - 2y$ with $x^{(0)} = \begin{bmatrix} 0 & 0 \end{bmatrix}^T$ and $H^{(0)} = I$.

Suppose that after one iteration, the next point is calculated as $x^{(1)} = \begin{bmatrix} 1 & 0 \end{bmatrix}^T$. What is $H^{(1)}$?

DFP update: $s^{(k)} = x^{(k+1)} - x^{(k)}$, $y^{(k)} = \nabla f(x^{(k+1)}) - \nabla f(x^{(k)})$,

$$H^{(k+1)} = H^{(k)} + \frac{s^{(k)}(s^{(k)})^T}{(s^{(k)})^T y^{(k)}} - \frac{H^{(k)} y^{(k)} (y^{(k)})^T H^{(k)}}{(y^{(k)})^T H^{(k)} y^{(k)}}.$$

a. $\begin{bmatrix} 2 & 1 \\ 1 & \frac{7}{4} \end{bmatrix}$

b. $\begin{bmatrix} \frac{6}{5} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{4}{5} \end{bmatrix}$

c. $\begin{bmatrix} \frac{7}{10} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{4}{5} \end{bmatrix}$

d. $\begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix}$

e. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$f(x, y) = x^2 + 2y^2 + xy - 4x - 2y$

$$f(x,y) = x^2 + 2y^2 + xy - 4x - 2y$$

$$x^{(0)} = (0, 0)^T$$

$$H^{(0)} = \underline{I}$$

$$x^{(1)} = (1, 0)^T$$

$$S^{(0)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\nabla f = \begin{bmatrix} 2x + y - 4 \\ 4y + x - 2 \end{bmatrix}$$

$$y^{(0)} = \begin{pmatrix} -2 \\ -1 \end{pmatrix} - \begin{pmatrix} -4 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\nabla f(x^{(0)}) = \begin{pmatrix} -4 \\ -2 \end{pmatrix}$$

Corrective term.

$$\nabla f(x^{(1)}) = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$$

$$\frac{\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \end{pmatrix}}{\begin{pmatrix} 1, 0 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix}} = \frac{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}}{2} = \begin{pmatrix} 1/2 & 0 \\ 0 & 0 \end{pmatrix}$$